

[Lecture Notebook] Formulating a Research Question

GE2021 Research Technology, Spring 2025

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Fri, 28 February 2025

Learning Objectives

	X	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
1	Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
2	Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
3	Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
4	Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
5	Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
6	Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

X	mpg	cyl	disp
Length:32	Min. :10.40	Min. :4.000	Min. : 71.1
Class :character	1st Qu.:15.43	1st Qu.:4.000	1st Qu.:120.8
Mode :character	Median :19.20	Median :6.000	Median :196.3
	Mean :20.09	Mean :6.188	Mean :230.7
	3rd Qu.:22.80	3rd Qu.:8.000	3rd Qu.:326.0
	Max. :33.90	Max. :8.000	Max. :472.0
hp	drat	wt	qsec
Min. : 52.0	Min. :2.760	Min. :1.513	Min. :14.50
1st Qu.: 96.5	1st Qu.:3.080	1st Qu.:2.581	1st Qu.:16.89
Median :123.0	Median :3.695	Median :3.325	Median :17.71
Mean :146.7	Mean :3.597	Mean :3.217	Mean :17.85
3rd Qu.:180.0	3rd Qu.:3.920	3rd Qu.:3.610	3rd Qu.:18.90
Max. :335.0	Max. :4.930	Max. :5.424	Max. :22.90
vs	am	gear	carb
Min. :0.0000	Min. :0.0000	Min. :3.000	Min. :1.000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:3.000	1st Qu.:2.000
Median :0.0000	Median :0.0000	Median :4.000	Median :2.000
Mean :0.4375	Mean :0.4062	Mean :3.688	Mean :2.812
3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:4.000	3rd Qu.:4.000
Max. :1.0000	Max. :1.0000	Max. :5.000	Max. :8.000

34 **Reference**